Top 20 Linux Administrator Interview Guide Real-World Q&A

By Ramesh Singh

Linux Administrator Interview Questions & Answers For Linux_DevOps _AWS_SysAdmin

Preparing for a Linux Admin role? Here are some practical LVM & User Management interview questions with clear answers 👇

🔹 Q1. How do you create an LV over a VG (with/without PE)?
👉 `lvcreate -L 5G -n lv_name vg_name` (size-based)
👉 `lvcreate -l 100%FREE -n lv_name vg_name` (PE-based)

🔹 Q2. How to format and mount an LV permanently?
👉 Format: `mkfs.ext4 /dev/vg/lv`
👉 Mount: `mount /dev/vg/lv /mnt/data`
👉 Permanent: `Add entry in /etc/fstab`

🔹 Q3. How to extend or reduce VG size?
👉 Extend: `vgextend vg_name /dev/sdX`
👉 Reduce: `vgreduce vg_name /dev/sdX`

🔹 Q4. How to extend LV and resize filesystem?
👉 `lvextend -L +5G /dev/vg/lv`
👉 `resize2fs /dev/vg/lv (for ext4)`

🔹 Q5. How to migrate LV data from a faulty PV?
👉 `pvmove /dev/sdX`

🔹 Q6. How to remove LV, VG, PV?
👉 `lvremove, vgremove, pvremove`

🔹 Q7. Types of users in Linux?
✔ Superuser (root)
✔ Regular user
✔ Service/System user

🔹 Q8. Which files store user details and passwords?
👉 `/etc/passwd` (user info)
👉 `/etc/shadow` (encrypted passwords)

🔹 Q9. User management commands?
👉 Add: `useradd`
👉 Modify: `usermod`
👉 Delete: `userdel -r`
👉 Change shell: `usermod -s /bin/bash username`

⬙ Q10. Lock and unlock a user?

☞ Lock: `passwd -l username`

☞ Unlock: `passwd -u username`

⬙ Q11. Force password change at first login?

☞ `chage -d 0 username`

⬙ Q12. Restrict login time using PAM?

☞ Configure `/etc/security/time.conf` with `pam_time.so`

⬙ Q13. Grant superuser privileges?

☞ Add user to wheel group or configure via `visudo`

⬙ Q14. Difference between `groups` and `id`?

☞ `groups` → shows group names

☞ `id → shows` UID, GID, and groups

⬙ Q15. Group management commands?

☞ `groupadd, groupmod, groupdel`

⬙ Q16. Remove a user from a group?

☞ `gpasswd -d user group`

⬙ Q17. Add user to secondary group without overwriting?

☞ `usermod -aG group user`

⬙ Q18. Use of `/etc/profile` and `/etc/login.defs`?

☞ `/etc/profile` → global environment settings

☞ `/etc/login.defs` → login policies

⬙ Q19. What are SUID, SGID, Sticky Bit?

✓ SUID → run as file owner

✓ SGID → run as group owner

✓ Sticky Bit → restrict deletion in shared directories

⬙ Q20. Set standard permissions?

☞ `chmod 755 file`

☞ `chown user:group file`